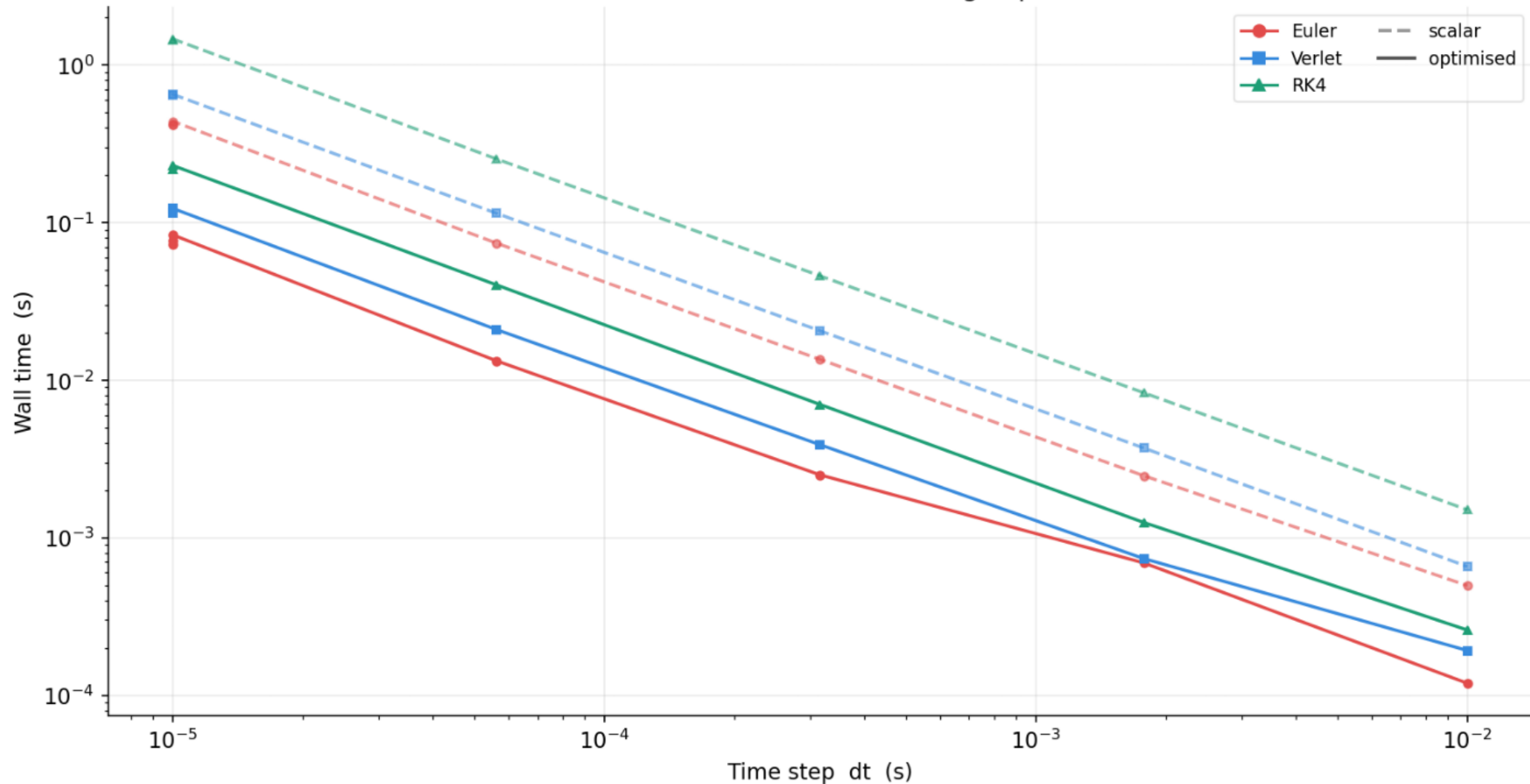
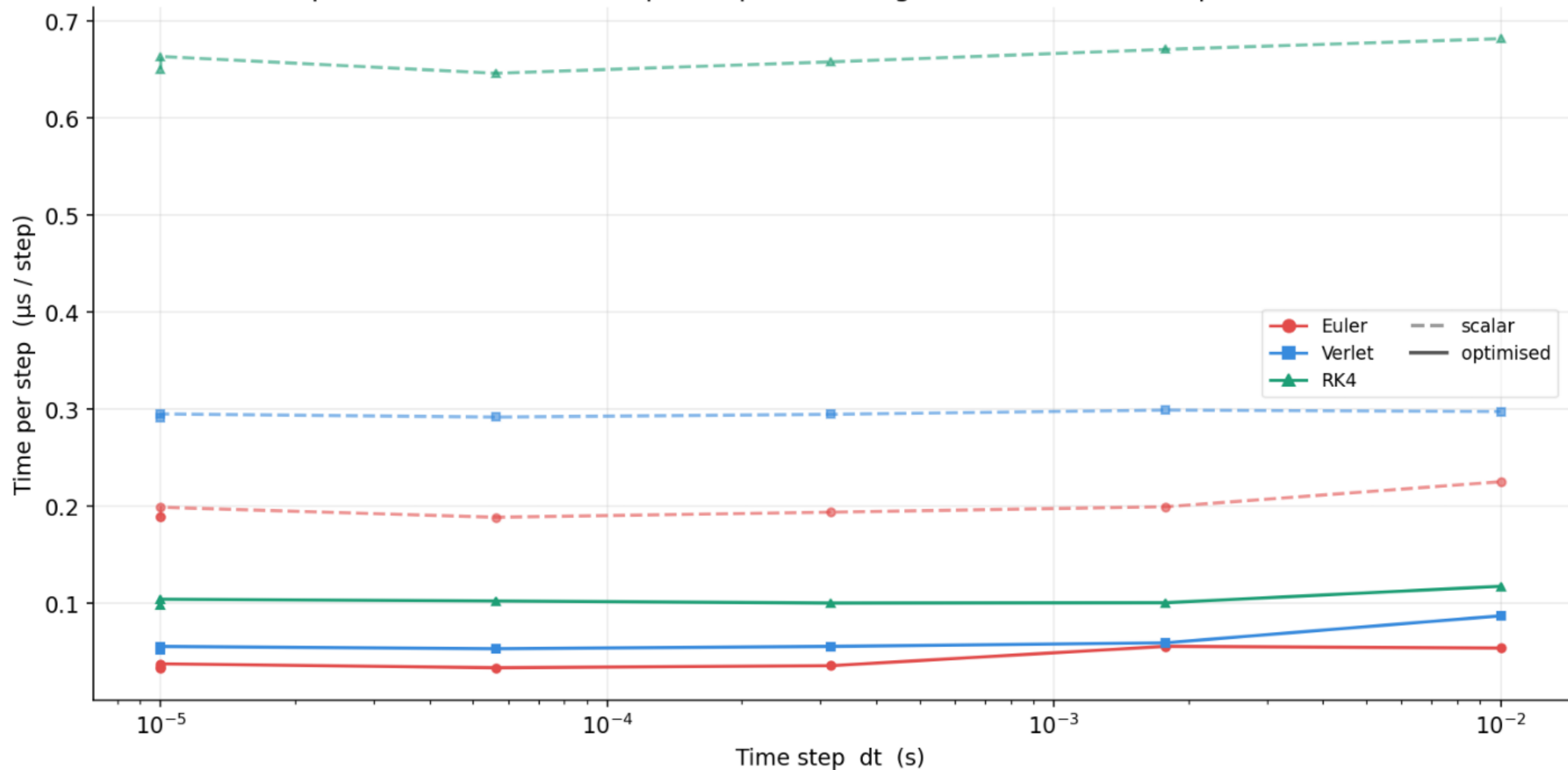


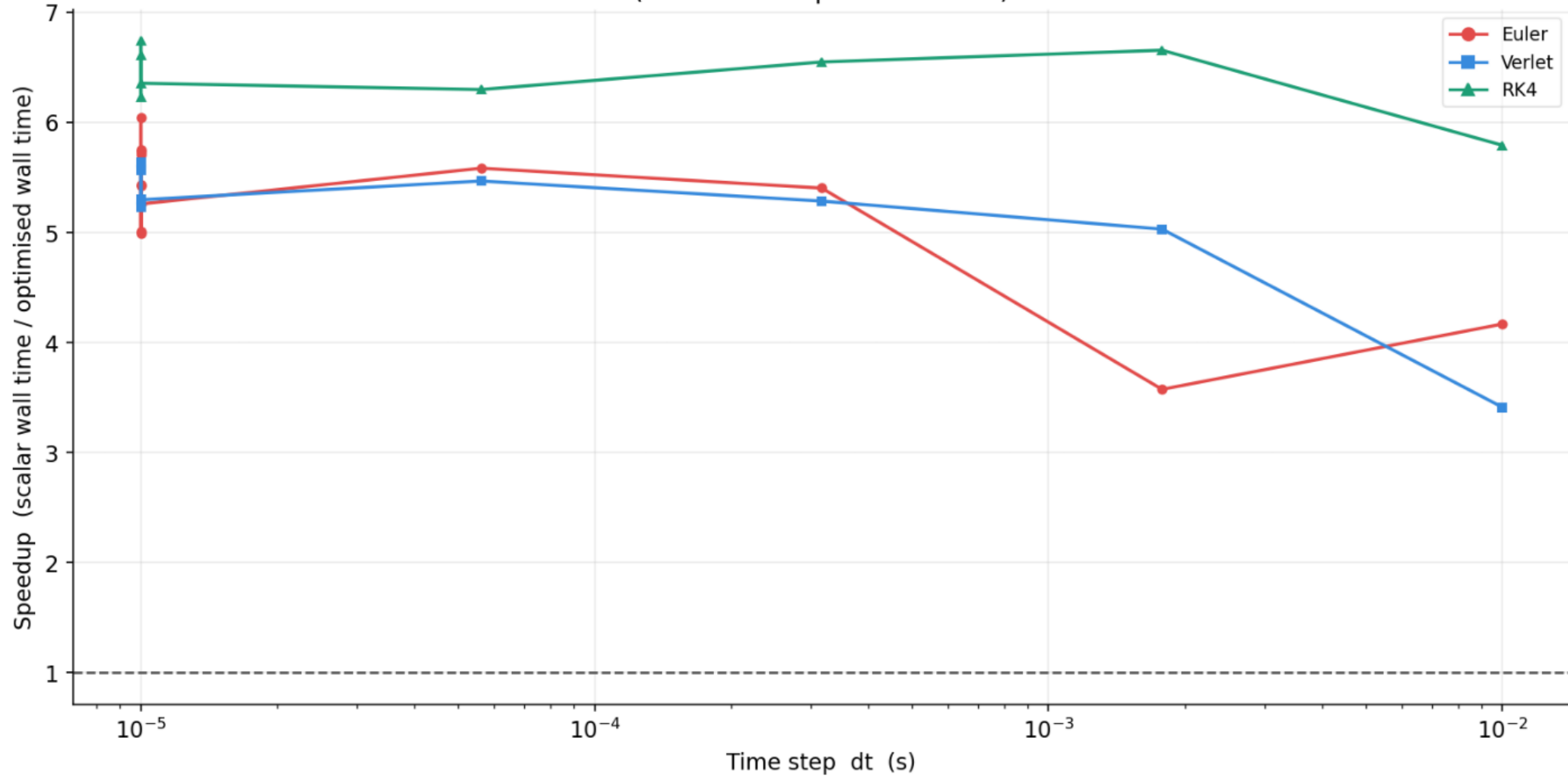
Wall time vs dt — $O(1/dt)$ scaling expected



Time per step vs dt
(plateau at fine dt = true per-step cost; rising at coarse dt = startup overhead)

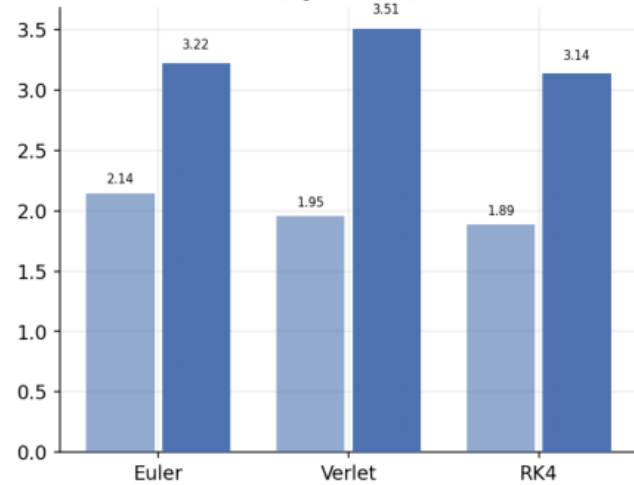


Scalar → Optimised speedup vs dt
(> 1 means optimised wins)

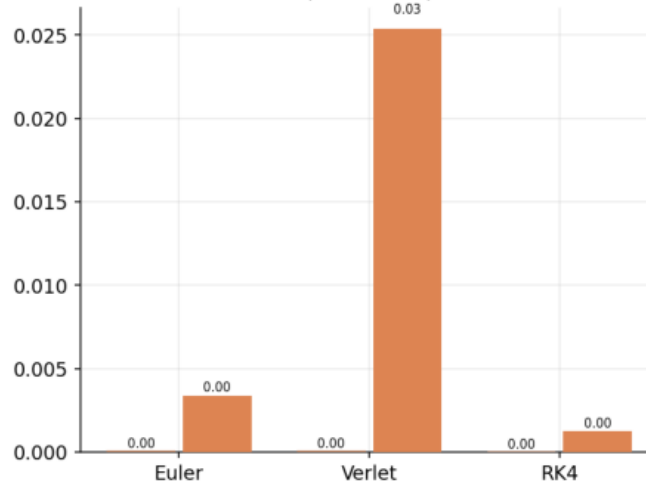


Hardware Counters — perf stat (grouped by solver, shading = profile)

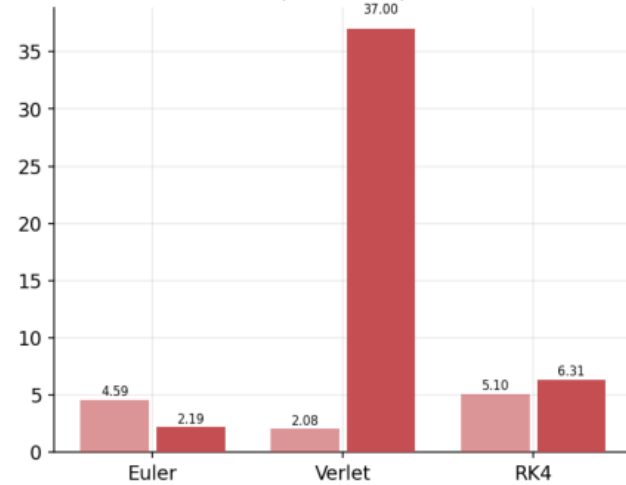
IPC
(higher is better)



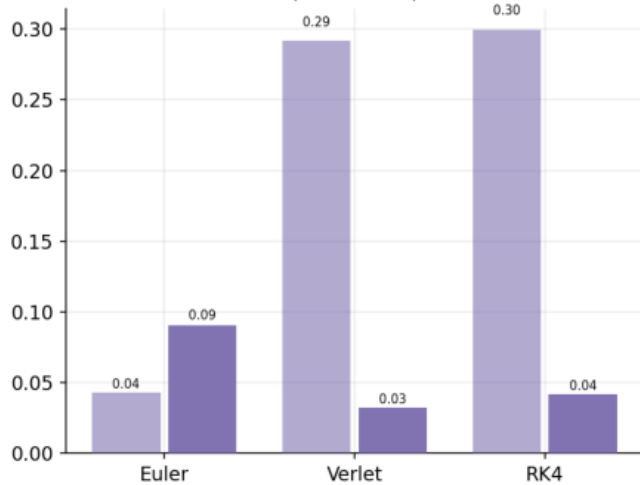
L1 miss %
(lower is better)



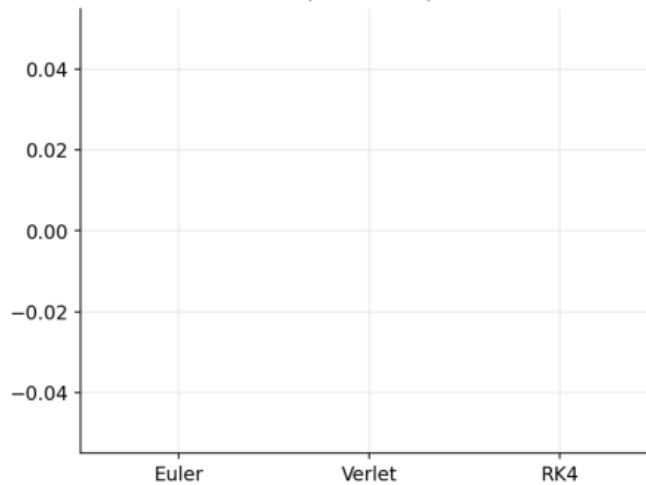
LLC miss %
(lower is better)



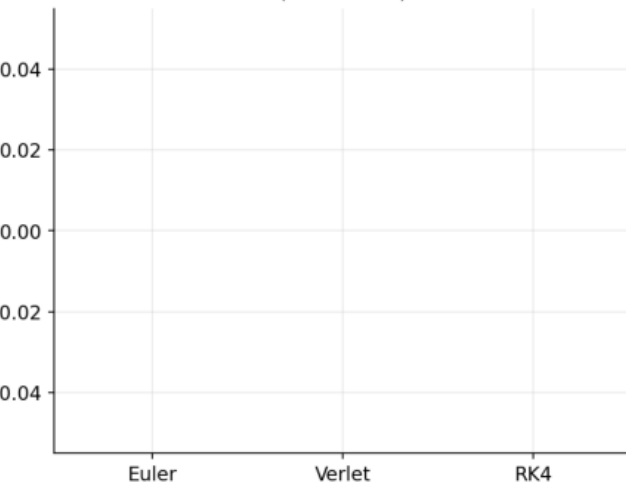
branch miss %
(lower is better)



stall fe %
(lower is better)

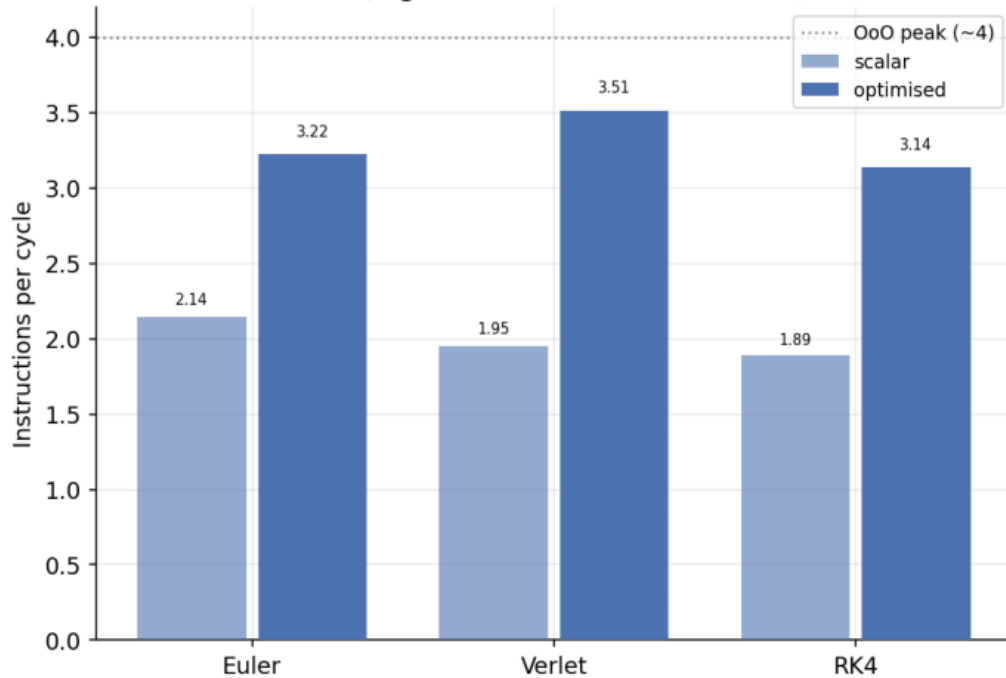


stall be %
(lower is better)

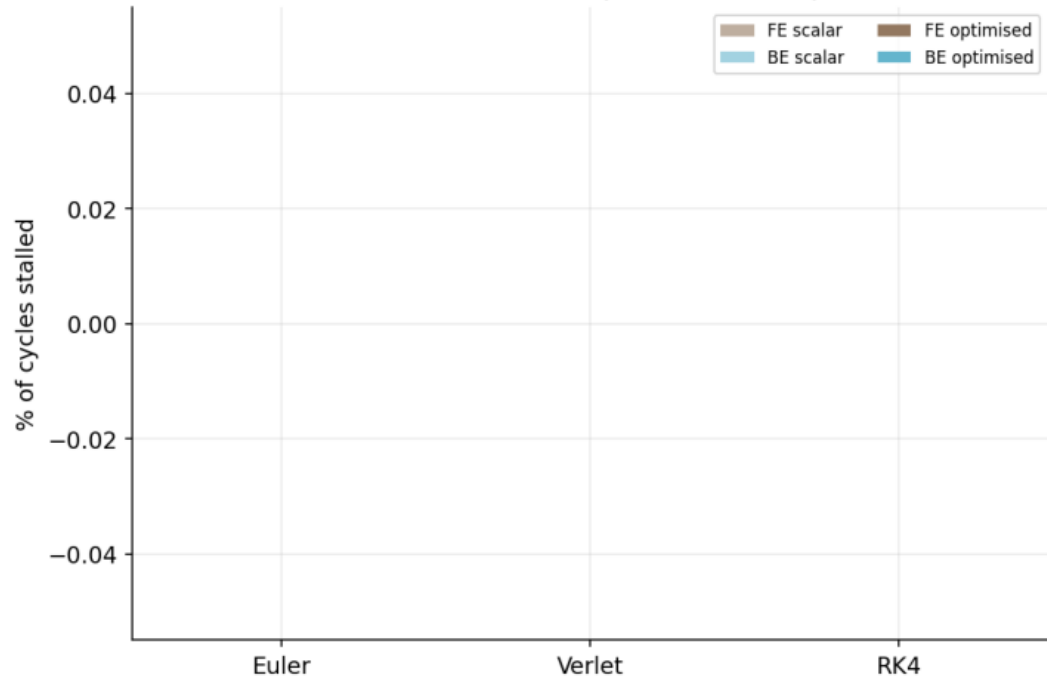


-- scalar — optimised

IPC (higher = better CPU utilisation)

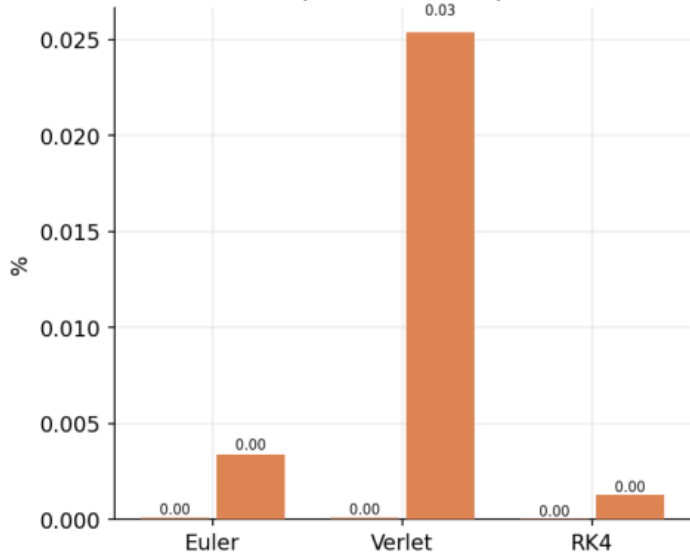


Stall breakdown (lower = better)

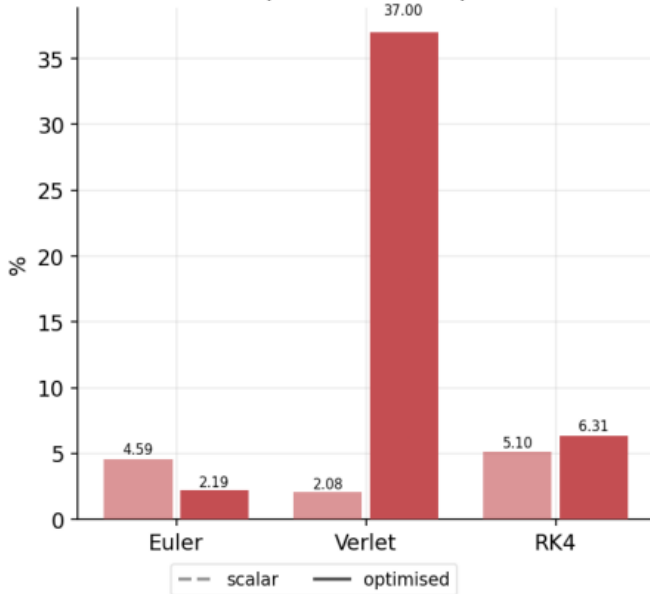


Cache & Branch Performance — perf stat

L1 D-cache miss rate (%)
(lower is better)



LLC miss rate (%)
(lower is better)



Branch miss rate (%)
(lower is better)

